THE JOURNAL OF PHYSICAL CHEMISTRY LETTERS
A JOURNAL OF THE AMERICAN CHEMICAL SOCIETY

pubs.acs.org/JPCL Letter

Interfacial Density-of-States Engineering Enables Transport Mechanism Transition in P3HT/Nitrogen-Doped Graphene Thermoelectric Composites

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Cite This: J. Phys. Chem. Lett. 2026, 17, 3565−3573 Read Online

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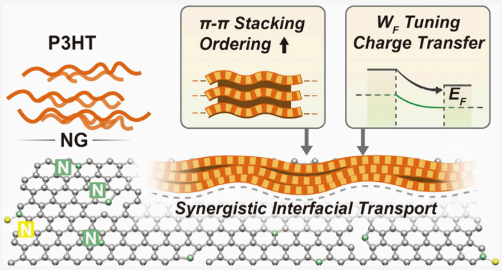
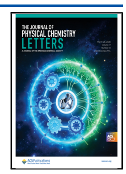
ABSTRACT: The achievable power factor of organic thermoelectric materials is constrained by thermoelectric transport mechanism coupled between the Seebeck coefficient (S) and electrical conductivity (σ). Here, we construct a nitrogen-doped graphene (NG)/poly(3-hexylthiophene) (P3HT) composite system, in which the thermoelectric transport mechanism can be continuously tuned through interfacial density-of-states (DOS) engineering between NG and P3HT. A direct interplay between polymer ordering, interfacial electronic structure, and carrier energy distribution was investigated. It is revealed that high P3HT concentration induces a Pisarenko-dominated regime with increased σ and suppressed thermopower; meanwhile, moderate thermal annealing generates NG-induced interfacial potential barriers, enabling energy-selective carrier transport that simultaneously enhances S and σ. Furthermore, excessive annealing time or temperature introduces DOS scattering associated with backbone disorder, leading to deteriorated transport performance. These results provide direct spectroscopic evidence for a transition from σ-controlled to energy-filtering-dominated transport in organic thermoelectric composites for decoupling thermoelectric parameters through interfacial electronic structure engineering.



Organic thermoelectric (TE) materials have attracted increasing attention owing to their intrinsic advantages, including low thermal conductivity, mechanical flexibility, lightweight nature, and solution processability, thereby enabling the applications in low-grade heat harvesting and wearable energy conversion devices.[1−6] Among them, conducting polymers such as poly(3-hexylthiophene) (P3HT) have been extensively studied because of their tunable electronic structure, controllable morphology, and compatibility with large-area fabrication.[7,8] However, the practical performance of organic TE remains fundamentally limited by the strong interdependence between the Seebeck coefficient (S) and electrical conductivity (σ), which is typically governed by the Pisarenko relationship.[9] According to this framework, an increase in carrier concentration enhances σ while inevitably suppresses S, leading to a long-standing trade-off that constrains the achievable power factor.[10]

To address the limitation, various strategies have been focused on modulating carrier generation, electron delocalization, and transport dimensionality beyond carrier-concentration control, including molecular doping,[3,11−14] all-conjugated polymer systems construction,[15] and nanocomposite engineering.[16−19] Molecular doping increases carrier concentration by shifting the Fermi level into the density-of-state (DOS); however, Coulomb binding between polarons, counterions and dopant-induced morphological perturbations often limit the fraction of mobile carriers.[11,12] All-conjugated donor−acceptor polymers enable ground-state charge transfer without counterions, promoting highly mobile interfacial transport with quasi-two-dimensional character.[15] Nanocomposite engineering introduces conductive fillers or extended π-systems that establish percolative pathways and locally modify the polymer electronic landscape.[16−18] In such systems, charge transport is no longer governed solely by thermally activated hopping within disordered polymer domains, but instead reflects a cooperative contribution from polymer ordering, interfacial charge transfer, and network-assisted transport. Improving the structural order of conjugated polymers effectively enhances carrier concentration through promoted π-π stacking and extended conjugation length.[20−22] Nevertheless, higher structural order or stronger doping often shifts the Fermi level deeper into the DOS, which increases carrier concentration at the expense of thermopower.[23] Recent studies suggest that interfacial engineering and DOS modulation provide alternative routes to decouple the two thermoelectric parameters.[24−28] In particular, energy filtering mechanisms, in which low-energy charge carriers are selectively scattered while

Received: February 11, 2026
Revised: March 5, 2026
Accepted: March 6, 2026
Published: March 11, 2026



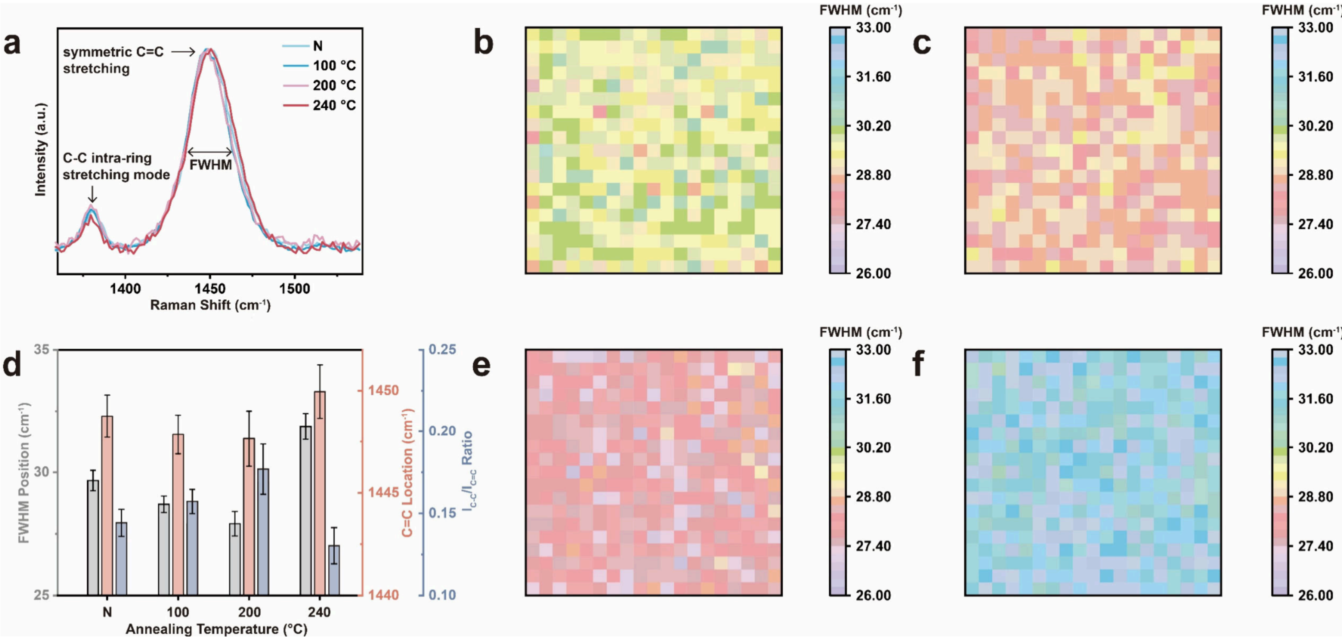


Figure 1. Raman characterization of the molecular ordering of P3HT at a fixed concentration of 15 mg mL^{-1} on NG under different annealing temperatures. (a) Raman spectra of P3HT/NG films under nonannealed (N), 100 °C, 200 °C, and 240 °C annealing conditions. Statistical heat maps of the fwhm of the P3HT C=C stretching mode extracted from Raman mapping measurements for P3HT/NG composite films under different annealing temperatures: (b) nonannealed (N), (c) 100 °C, (e) 200 °C, and (f) 240 °C. Each pixel represents the fwhm value obtained from an individual Raman spectrum collected within the mapped area. The color contrast reflects the statistical distribution of fwhm values of the films. (d) Statistic diagram of the fwhm of the P3HT C=C stretching mode, C=C location, and $I_{C-C}/I_{C=C}$ ratios extracted from Raman mapping measurements for P3HT/NG composite films under different annealing temperatures.

high-energy carriers dominate transport, have been proposed to effectively enhance S without severely compromising σ.[29−31] Such mechanisms are frequently realized in inorganic nanocomposites through potential barriers or heterointerfaces.[32−34] However, in organic TE systems, direct experimental evidence linking interfacial electronic structure, DOS distribution, and transport mechanism transitions remains scarce.[35]

Therefore, developing a organic TE model system which directly correlates with these attributes is of critical importance.[36] Graphene, owing to its atomically thin structure, strong $\pi-\pi$ interaction, and tunable Fermi level, enables modulation of interfacial charge distribution and DOS. Therefore, graphene/P3HT heterostructures can simultaneously tune polymer ordering, interfacial electronic structure, and carrier energy distribution.[37,38] For example, Skrypnychuk et al. constructed well-controlled P3HT/monolayer graphene heterostructures and demonstrated that graphene significantly alters the crystallite orientation of P3HT, promoting mixed face-on and edge-on lamellar stacking and substantially enhancing vertical charge transport.[39] Similar graphene/P3HT systems have also been explored in organic field-effect transistors and photovoltaic devices, where graphene films primarily function as transparent electrodes or tunable Schottky contacts to modulate charge injection and isothermal electrical transport.[40−42] Although enhanced σ have been reported, the underlying mechanism is generally attributed to conductivity improvement rather than interface-induced modulation of carrier energetics; especially, these works primarily do not address how graphene-induced interfacial electronic structure modulation influences thermoelectric transport, particularly the intrinsic coupling between the S and σ. Alternatively, nitrogen-doping into graphene introduces the localized electronic states, which can further modulate the interfacial DOS in the graphene-based heterostructures and potentially enable energy-selective carrier transport.[43] Notably, a systematic analysis on how electronic coupling between N-doped graphene (NG) and P3HT reshapes the interfacial DOS and regulates the transport regime remains largely unexplored.

In this work, we systematically investigate the carrier and thermoelectric transport mechanisms in the P3HT/NG composite films by jointly tuning P3HT concentration and thermal annealing conditions. A comprehensive experimental approach is employed to establish a direct correlation between structural order, electronic structure, and transport properties. By correlating these electronic structure parameters with thermoelectric properties, we demonstrate a clear transition from Pisarenko-dominated transport at high polymer concentration to an energy-filtering-dominated regime induced by moderate thermal annealing. Furthermore, excessive annealing time or temperature is shown to introduce DOS scattering associated with backbone disorder, leading to deteriorated transport performance. Unlike previously reported transport transitions mainly driven by bulk structural evolution or molecular ordering, the present study explores an interface-dominated mechanism based on the DOS modulation at the P3HT/N-doped graphene interface. This study elucidates the interplay between polymer ordering, interfacial electronic structure, and thermoelectric transport, offering a mechanistic framework for breaking the conventional S-σ trade-off in the organic TE composites.

To establish a foundation for the analysis of interfacial electronic effects, the growth parameters of NG were optimized to ensure the film continuity and N doping. The relevant characterization data are given in Figures S1−S5 and Table S1. From these data, the optimized growth parameters are determined as follows: growth temperature of 1030 °C, N

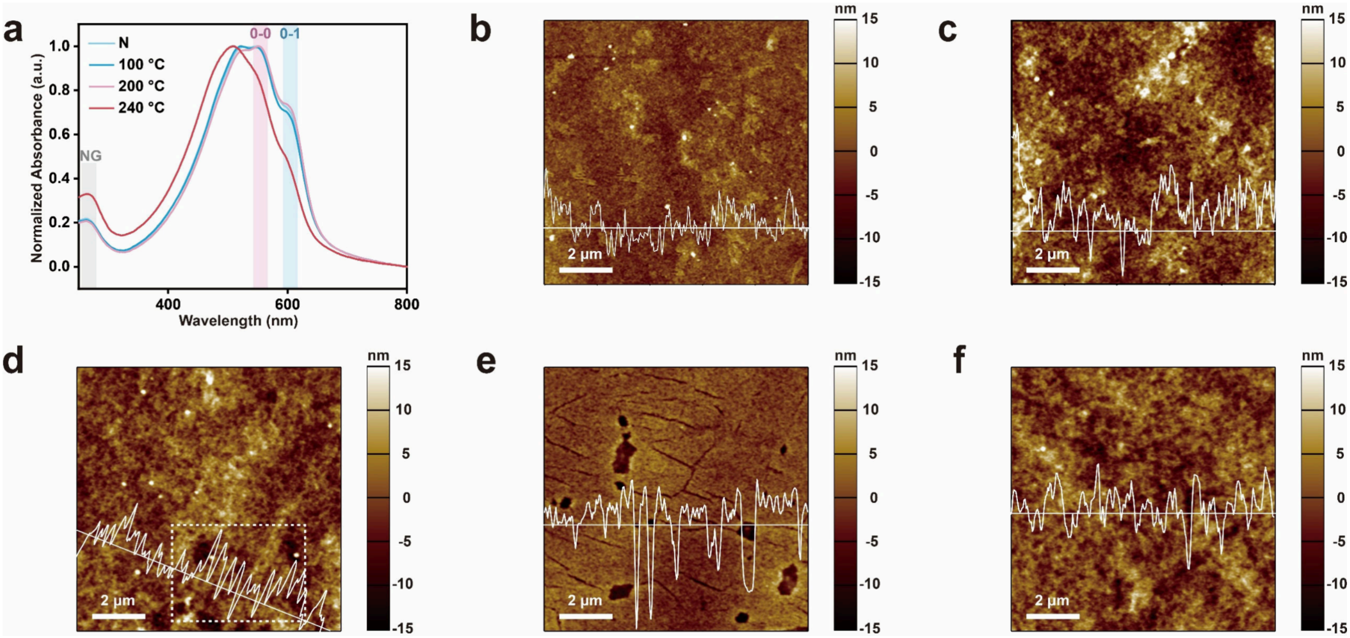


Figure 2. (a) UV−vis absorption spectra of P3HT/NG composite films with a fixed P3HT concentration of 15 mg mL^{-1} under different annealing temperatures (nonannealed (N), 100 °C, 200 °C, and 240 °C). (b−f) AFM images of P3HT films under different processing conditions: (b) P3HT concentration of 5 mg mL^{-1} without annealing, (c) P3HT concentration of 15 mg mL^{-1} without annealing, (d) 15 mg mL^{-1} P3HT annealed at 200 °C for 20 min, (e) 15 mg mL^{-1} P3HT annealed at 240 °C for 20 min, and (f) 15 mg mL^{-1} P3HT annealed at 200 °C for 60 min. All images were acquired in tapping mode over a scan area of 10 × 10 μm^2.

introducing time at the last 3 min. The intensity ratios of the Raman 2D to G bands (I_{2D}/I_G) are 2.11 for pristine graphene and 1.71 for NG, respectively, indicating the monolayer nature of both samples. The reduced I_{2D}/I_G ratio and the increased fwhm after N doping suggests an increased electronic perturbation and lattice disorder induced by heteroatom incorporation, while preserving the overall graphene framework. Optical microscopy (OM) images indicate that the NG films exhibit a relatively uniform thickness over a large area. X-ray photoelectron spectroscopy (XPS) analysis reveals a nitrogen doping level of 5.13%, with pyrrolic-N, graphitic-N, and oxidized-N identified as the dominant nitrogen configurations.[44] Graphitic-N is generally associated with substitutional doping and Fermi level shift, whereas pyrrolic-N and oxidized-N tend to introduce localized electronic states and structural perturbations. Compared with pristine graphene, N doping introduces localized electronic states and modifies the Fermi level, thereby enabling active modulation of the interfacial DOS in hybrid systems.[43,44]

As shown in Figure S6, the characteristic Raman vibrational modes of P3HT are observed at approximately 728, 1180, 1208, 1381, and 1452 cm^{-1}, consistent with previous reports. No discernible spectral shifts or additional features are detected after deposition on NG, indicating that the chemical structure of P3HT remains intact. The Raman signatures of graphene are not clearly resolved due to the overlap between the graphene G and D bands and the strong P3HT modes at 1381 and 1452 cm^{-1}, as well as the dominant Raman signal from the comparatively thicker P3HT layer.[38] Figure 1 and Figures S7−S9 present a systematic Raman spectroscopic investigation of the molecular ordering and stacking behavior of P3HT on NG under orthogonally designed processing conditions, including polymer concentration and thermal annealing temperature. Raman spectroscopy is a sensitive probe for the backbone conformation and crystalline ordering of P3HT, particularly through the analysis of two characteristic in-plane ring skeleton vibrational modes.[20] As shown in Figure 1a, all spectra exhibit two characteristic peaks centered at approximately 1455 cm^{-1}, corresponding to the symmetric C=C stretching mode, and around 1381 cm^{-1}, assigned to the C−C intraring stretching mode. The relative evolution of these modes provides direct insight into the ordering degree of P3HT phase: (i) the peak position of the symmetric C=C stretching mode, (ii) the full width at half-maximum (fwhm) of C=C mode, and (iii) the intensity ratio between the C−C and C=C modes ($I_{C-C}/I_{C=C}$). A more ordered P3HT phase is typically characterized by a slight redshift of the C=C peak, a narrower fwhm, and an increased $I_{C-C}/I_{C=C}$ ratio, reflecting enhanced intrachain planarity and interchain coupling.[45] Under a fixed P3HT concentration of 15 mg mL^{-1}, systematic thermal annealing gives rise to a pronounced and continuous evolution of these Raman features, as summarized in Figure 1b-f and Figure S4. Specifically, the nonannealed sample illustrates large fwhm. In comparison, annealing at 100 °C already leads to discernible signatures of enhanced molecular ordering, including a slightly narrowed C=C stretching mode and an increased $I_{C-C}/I_{C=C}$ ratio. This indicates that even mild thermal treatment promotes partial backbone planarization and interchain packing of P3HT on NG. However, the extent of structural reorganization is limited by the relatively low annealing temperature, resulting in only moderate improvement in ordering. Notably, annealing at 200 °C results in the narrowest C=C peak fwhm and the highest intensity ratio, suggesting that this temperature is optimal for promoting P3HT ordering on the NG surface. In contrast, further increasing the annealing temperature to 240 °C leads to a broadening of the C=C peak and a reduction in the intensity ratio, implying thermally induced disorder or disruption of conjugated segments. This degradation of Raman ordering parameters suggests that excessive thermal energy disrupts the

https://doi.org/10.1021/acs.jpclett.6c00487
J. Phys. Chem. Lett. 2026, 17, 3565−3573

conjugated backbone integrity or induces local disorder, counteracting the ordering benefits achieved at intermediate temperatures.

Figure 2a presents the UV−vis absorption spectra of P3HT/NG composite films with a fixed P3HT concentration of 15 mg mL^{-1} under different annealing temperatures. A distinct absorption band around 264 nm is observed for all samples confirming the presence and optical contribution of the NG framework in the composite films.[46] Two pronounced absorption features centered at approximately 605 and 550 nm are observed, which are assigned to the 0−0 and 0−1 vibronic transitions of P3HT, respectively. According to the weakly coupled H-aggregation model proposed by Spano,[47] the relative intensity of the 0−0 to 0−1 transitions (I_{0-0}/I_{0-1}) serves as a sensitive indicator of the balance between intrachain and interchain interactions in conjugated polymers. An increased I_{0-0}/I_{0-1} ratio generally reflects enhanced intrachain ordering, reduced energetic disorder, and increased J-aggregation character associated with improved backbone planarization and extended conjugation length. Compared with the nonannealed sample, mild thermal annealing at 100 °C leads to a moderate enhancement of the 0−0 transition, suggesting partial relaxation of chain conformation and incipient ordering of P3HT on the NG surface. Upon annealing at 200 °C, the 0−0 peak intensity is maximized and accompanied by a subtle red-shift of the vibronic structure, indicating the highest degree of intrachain ordering and backbone planarization. In sharp contrast, annealing at 240 °C results in a pronounced loss of vibronic definition: the characteristic 0−0 and 0−1 transitions become significantly broadened and less distinguishable, indicating that excessive thermal energy disrupts the conjugated backbone and suppresses coherent vibronic coupling. This degradation of spectral features directly reflects thermally induced conformational disorder in P3HT. It is noted that the annealing process was conducted under an inert Ar atmosphere, under which P3HT is generally thermally stable for short processing durations. The absence of new Raman features together with the preserved characteristic UV−vis absorption profile suggests that the observed changes originate primarily from loss of structural ordering rather than chemical degradation of the polymer backbone or covalent bond formation with NG.

As illustrated in Figures S10 and S11, the concentration-dependent UV−vis spectra reveal a monotonic enhancement of the 0−0 transition with increasing P3HT concentration under identical thermal conditions, implying progressively improved intrachain interactions and molecular ordering on NG. Meanwhile, the annealing time-dependent spectra demonstrate that a moderate annealing duration optimizes vibronic definition, whereas prolonged annealing leads to spectral broadening, consistent with the onset of thermal disorder. Importantly, the optimal concentration, annealing temperature, and annealing time extracted from UV−vis analysis, are fully consistent with the Raman results, collectively confirming that controlled thermal treatment and polymer loading synergistically regulate P3HT ordering on the NG surface.

Complementary Raman line scans and statistical analyses for P3HT concentrations of 5, 10, and 20 mg mL^{-1} are provided in Figures S12 and S13. Across all annealing temperatures, higher concentrations consistently lead to reduced fwhm and enhanced $I_{C-C}/I_{C=C}$ ratios, indicating a concentration-driven improvement in molecular ordering on NG. Notably, the optimal annealing temperature of 200 °C remains valid across different concentrations, demonstrating the robustness of the identified ordering mechanism. Collectively, these Raman results establish 200 °C as the optimal annealing condition and confirm that both concentration and thermal treatment synergistically govern the ordering and stacking of P3HT on NG.

To further elucidate the effect of annealing duration, a single-factor investigation was performed at a fixed concentration of 15 mg mL^{-1} and an annealing temperature of 200 °C, with annealing times of 40 and 60 min. Raman mapping images of samples annealed for 40 and 60 min are shown in Figure S14, while the corresponding Raman spectra and statistical analysis of the three ordering-related parameters are presented in Figures S10d and S11d. The results demonstrate that annealing for 20 min significantly enhances P3HT ordering compared to the nonannealed film, whereas further prolonging the annealing time leads to a gradual deterioration of molecular order. This nonmonotonic behavior suggests that excessive thermal exposure may induce chain scission, conformational disorder, or interfacial relaxation effects, counteracting the ordering benefits achieved at shorter annealing times.

To further elucidate how molecular ordering translates into mesoscale morphology, AFM was employed to investigate the surface topography of representative P3HT/NG composite films under different processing conditions. As shown in Figure 2b, the P3HT film prepared with a low concentration of 5 mg mL^{-1} exhibits a relatively smooth but featureless surface morphology, with weak height contrast and no discernible interconnected structures. Increasing the P3HT concentration to 15 mg mL^{-1} (Figure 2c) leads to a more heterogeneous surface with enhanced nanoscale contrast, suggesting the formation of denser polymer packing and improved film continuity. Although no long-range fibrillar features are observed, the increased surface roughness and domain contrast are indicative of enhanced π-π stacking, in agreement with the improved ordering inferred from Raman and UV−vis results. However, this concentration-induced densification primarily increases carrier concentration without introducing significant interfacial energy modulation, consistent with the persistence of Pisarenko-like transport behavior. Upon annealing the 15 mg mL^{-1} sample at 200 °C for 20 min (Figure 2d), the surface morphology undergoes a pronounced transformation. The AFM image reveals a cleaner and more homogeneous surface with reduced isolated features and enhanced domain connectivity. Notably, weakly elongated and interconnected textures begin to emerge, which can be regarded as an incipient fibrillar morphology (dashed box), indicating the onset of polymer chain aggregation and alignment without the formation of fully developed long-range fibrillar networks. This initial fibrillation reflects optimized chain rearrangement and enhanced interchain coupling, consistent with the highest molecular ordering identified by Raman line width narrowing and the strongest vibronic structure in UV−vis spectra. Such a morphology is highly favorable for coherent charge transport while simultaneously facilitating interfacial potential modulation at the P3HT/NG interface. In contrast, annealing at an elevated temperature of 240 °C results in pronounced morphological degradation, as shown in Figure 2e. The surface becomes less uniform, with irregular features and disrupted continuity, indicating partial thermal damage and backbone disorder in P3HT. This morphological deterioration is

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J. Phys. Chem. Lett. 2026, 17, 3565−3573

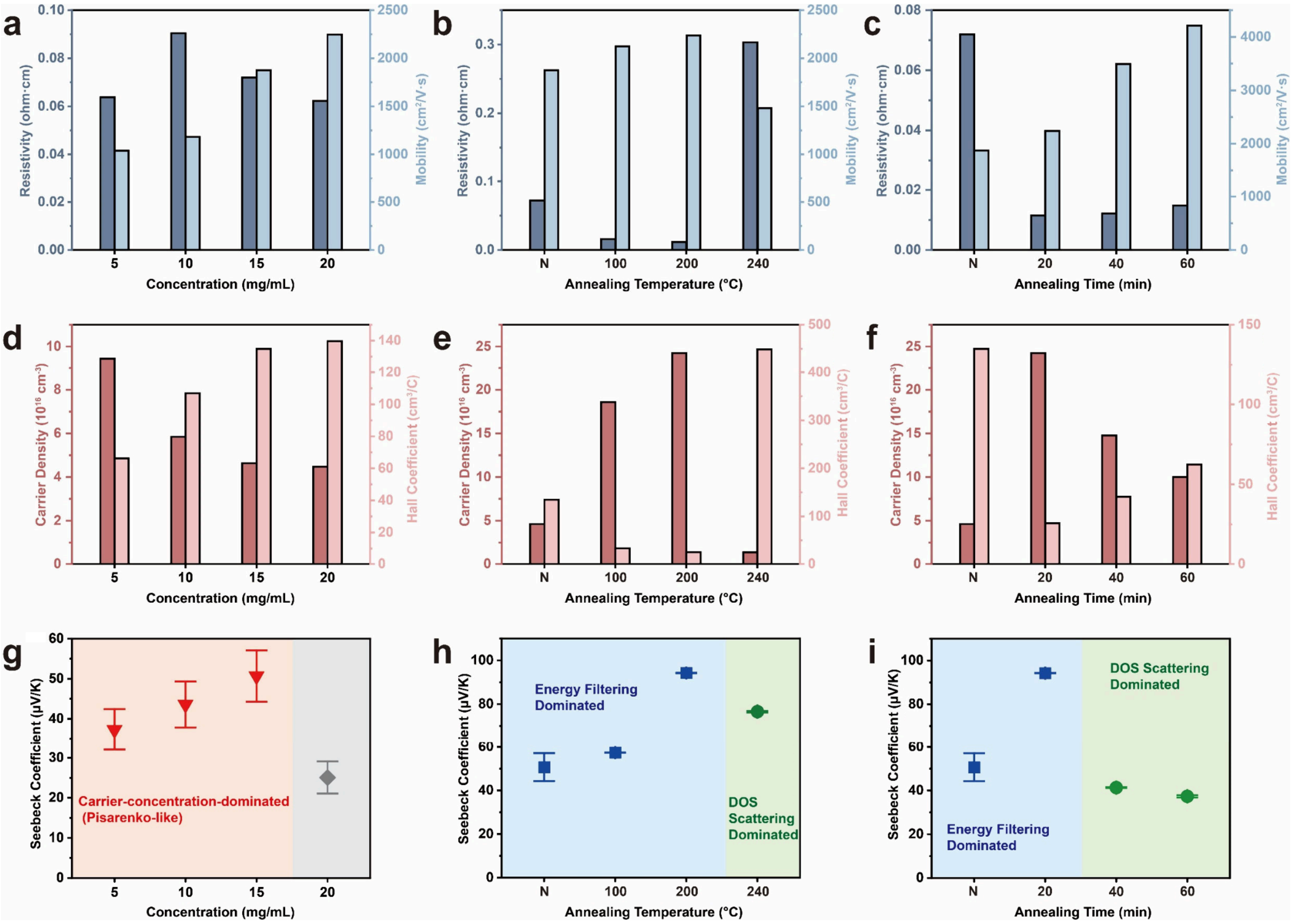

Figure 3. Charge transport and thermoelectric properties of P3HT/NG composite films under different processing conditions. Electrical resistivity and Hall mobility as a function of P3HT concentration (a), annealing temperature (b), and annealing time (c) and corresponding carrier concentration and Hall coefficient extracted from Hall measurements under the same conditions (d−f). Seebeck coefficient as a function of P3HT concentration (g), annealing temperature (h), and annealing time (i).

consistent with the reduced ordering observed spectroscopically. Similarly, extending the annealing time to 60 min at 200 °C (Figure 2f) leads to a loss of the optimized morphology observed at 20 min. Although the film remains continuous, the surface exhibits increased roughness and less well-defined domains, indicative of over-relaxation and partial disordering of polymer chains. This morphological evolution aligns with the decrease in molecular ordering detected by Raman and UV−vis analyses. The AFM results provide a crucial morphological bridge between the spectroscopic ordering information obtained from Raman and UV−vis analyses and the transport behaviors discussed.

The evolution of charge transport and thermoelectric properties of P3HT/NG composite films were investigated as the functions of P3HT concentration, annealing temperature, and annealing time. By systematically correlating carrier concentration (n), mobility (μ), resistivity (ρ), Hall coefficient (R_H), and Seebeck coefficient (S), distinct transport regimes can be identified under different processing conditions. Here, these parameters are related through $\sigma = ne\mu$, where e is the elementary charge. In the conventional thermoelectric framework, the S is primarily governed by n, whereas μ contributes to σ indirectly.

As shown in Figures 3a and 3d, increasing the P3HT concentration from 5 to 20 mg mL^{-1} leads to a pronouncedly increased n accompanied by a moderate variation in μ, resulting in a systematic decrease in ρ. Correspondingly, in the window of P3HT concentration from 5 to 15 mg mL^{-1}, the S value exhibits a monotonic reduction with increasing n (Figure 3g), which is characteristic of a carrier-concentration-dominated transport regime. This inverse relationship between S and n is consistent with the classical Pisarenko behavior commonly observed in doped conjugated polymers, where the upward shift of the Fermi level into a higher DOS region enhances electrical conductivity at the expense of thermopower.[10] Notably, despite the improved molecular ordering at higher concentrations as evidenced by Raman and UV−vis analyses, the thermoelectric response remains constrained by the intrinsic S−n coupling. This suggests that concentration-induced ordering primarily increases the DOS filling rather than generating interfacial energy modulation, and is insufficient to decouple S and σ in the P3HT/NG system. At the highest concentration of 20 mg mL^{-1}, the further increase in carrier concentration shifts the Fermi level deeper into the DOS, reducing the average transport energy of charge carriers and thereby leading to a pronounced decrease in the S.

In contrast, a distinctly different trend is observed upon tuning the annealing temperature. Compared with the nonannealed sample, thermal treatment at 100 °C already induces noticeable changes in the electronic transport

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J. Phys. Chem. Lett. 2026, 17, 3565−3573

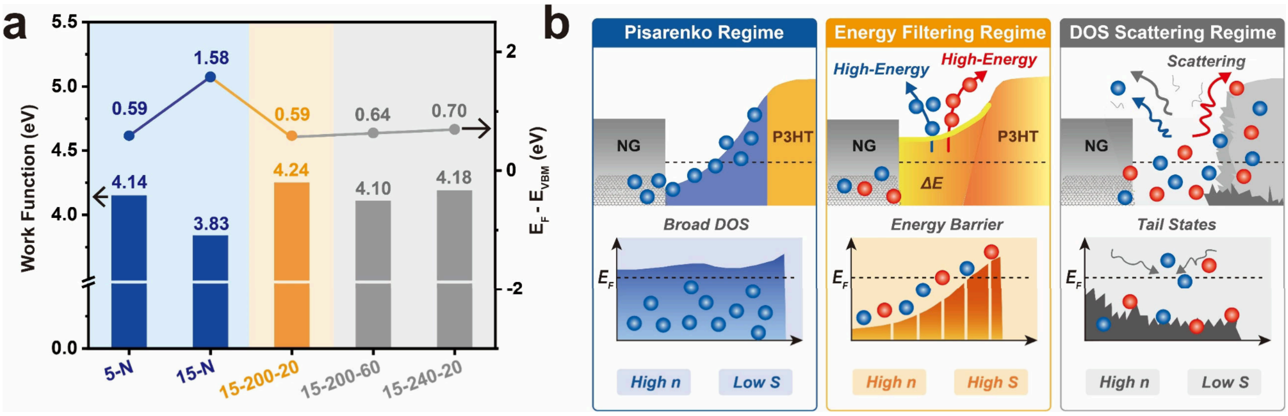

Figure 4. (a) Ultraviolet photoelectron spectroscopy (UPS) analysis of the P3HT/NG composite film. (left) Secondary electron cutoff (SECO) region used to determine the work function of the composite film, where the work function is extracted from the difference between the incident photon energy and the SECO onset. (right) Valence band spectrum near the Fermi level, with the valence band maximum (VBM) onset determined by linear extrapolation of the leading edge. (b) Schematic illustration of transport mechanism transition in P3HT/NG heterostructures. The red balls represent holes, and the blue balls represent electrons.

behavior. As illustrated in Figure 3b, annealing at 100 °C leads to a clear reduction in ρ accompanied by an increase in μ, indicating partial relaxation of polymer chain conformation and improved charge transport pathways at the P3HT/NG interface.[22] Meanwhile, the carrier density extracted from Hall measurements (Figure 3e) increases moderately relative to the nonannealed state, suggesting enhanced charge transfer or activation of additional electronic states upon mild thermal annealing. Importantly, the S value does not decrease with the increased carrier concentration at 100 °C; instead, a slight enhancement of S is observed (Figure 3h). This weak deviation from Pisarenko behavior indicates that the onset of interfacial energy modulation begins once P3HT chains become more ordered to better couple with the NG, although the effect remains limited. At this stage, the transport behavior can be regarded as an intermediate regime, where improved structural ordering contributes to better conductivity while the thermopower is not yet significantly amplified.

Upon increasing the annealing temperature to 200 °C, both polymer ordering and interfacial electronic coupling reach an optimal state, leading to a pronounced decrease in resistivity. More strikingly, the S value reaches its maximum value despite the elevated n, suggesting a prominent deviation from conventional Pisarenko-dominated transport. Such behavior is commonly associated with an energy-filtering-assisted transport mechanism, in which interfacial potential fluctuations selectively suppress low-energy carriers while enhancing the contribution of high-energy carriers to thermoelectric transport.[30] Similar phenomena have been reported in polymer-based thermoelectric systems incorporating low-dimensional fillers or forming strong interfacial electronic coupling, where interface-induced potential barriers and local DOS modulation give rise to an apparent decoupling between S and n.[48,49] In the present system, the coexistence of highly ordered P3HT domains and NG-induced interfacial electronic heterogeneity is therefore likely to reshape the local DOS, promoting selective high-energy carrier transport.

The emergence of this regime correlates well with the highest molecular ordering of P3HT at 200 °C, as revealed by Raman line width narrowing and enhanced vibronic features in UV−vis spectra. The improved ordering facilitates more coherent charge transport while enabling interfacial potential fluctuations at the P3HT/NG interface to selectively filter the carriers. However, further increasing the annealing temperature to 240 °C leads to a suppression of S despite a continued increase in n. Although the n remains relatively high, the S value decreases, signalizing the onset of DOS-scattering-dominated transport likely associated with excessive thermal disorder and partial disruption of the conjugated backbone. In this case, the loss of polymer ordering broadens the DOS near E_F, weakening energy filtering and increasing carrier scattering. Together, these results demonstrate a progressive evolution of transport mechanisms from a weakly modified Pisarenko-like regime to an optimized energy-filtering regime below 200 °C, followed by disorder-induced deterioration at higher annealing temperatures.

A similar nonmonotonic trend is observed when varying the annealing time at a fixed temperature of 200 °C (Figure 3c,f,i). Annealing for 20 min yields the highest S value together with elevated n consistent with the energy-filtering-dominated regime. Prolonged annealing beyond 20 min results in a gradual decline of S, accompanied by increased n and reduced Hall coefficient, indicating a transition toward a DOS-scattering-dominated transport regime. Despite the relatively high carrier mobility retained at longer annealing times, excessive thermal exposure induces partial disordering of P3HT chains, as supported by Raman and UV−vis analyses. This disorder diminishes interfacial DOS modulation and broadens the transport energy window, leading to enhanced carrier scattering and diminished thermopower. These results highlight the critical role of thermal processing time in balancing structural ordering and electronic disorder, and further emphasize that optimized thermoelectric performance arises from a narrow processing window.

Figure 4 summarizes the evolution of interfacial electronic structure in representative P3HT/NG samples and correlates it with the corresponding thermoelectric transport mechanisms. The reduced σ and increased n of NG relative to pristine graphene indicate N-doping induced interfacial electronic states, which provides the prerequisite interfacial potential fluctuations for different transport mechanisms. As shown in Figure 4a, both the work function (W_F) and the valence band maximum onset relative to the Fermi level ($E_F - E_{VBM}$) exhibit systematic variations that closely correlate with the Hall and

https://doi.org/10.1021/acs.jpclett.6c00487
J. Phys. Chem. Lett. 2026, 17, 3565−3573

Seebeck measurements discussed above. For the nonannealed sample, as shown in Figures S15 and S16, increasing the P3HT concentration from 5 mg mL^{-1} to 15 mg mL^{-1} results in a pronounced decrease in the W_F from 4.14 to 3.83 eV, accompanied by a substantial increase in $E_F - E_{VBM}$ from 0.59 to 1.58 eV. This indicates a downward shift of the Fermi level toward deeper valence-band states, consistent with a higher hole concentration and enhanced doping level. Correspondingly, Hall measurements reveal an increased n and reduced S, characteristic of Pisarenko-dominated transport, where thermopower decreases monotonically with increasing n. As illustrated in the left panel of Figure 4b, the UPS results confirm that the S is primarily governed by the position of the Fermi level within the DOS, rather than by interfacial energy modulation.

Upon annealing the 15 mg mL^{-1} sample at 200 °C for 20 min, the electronic structure undergoes a distinct transformation (Figure S17). The W_F increases markedly to 4.24 eV, while $E_F - E_{VBM}$ decreases to 0.59 eV, approaching the value observed in the low-concentration sample. Notably, this upward shift of W_F occurs simultaneously with an increase in both n and S, which deviates from the conventional Pisarenko relation. This apparent contradiction can be rationalized by an energy-filtering-dominated transport mechanism, where interfacial potential modulation at the P3HT/NG junction selectively suppresses low-energy carriers while favoring the transport of higher-energy carriers. The relatively smooth valence-band edge further supports the presence of an interfacial energy barrier rather than strong DOS broadening, as depicted in the central panel of Figure 4b. Combined with previous results showing the highest P3HT ordering at this condition, these findings suggest that optimized polymer ordering enhances interfacial coupling of P3HT with NG, enabling effective energy filtering without sacrificing carrier concentration.

Extending the annealing time to 60 min at 200 °C leads to a moderate reduction of W_F (4.10 eV) and a slight increase in $E_F - E_{VBM}$ to 0.70 eV. Although the valence-band edge remains relatively smooth (Figure S18), the diminished S and reduced n indicate that excessive annealing disrupts the optimal interfacial configuration. This behavior is consistent with partial loss of polymer ordering, as evidenced by Raman and UV−vis analyses, which weakens the energy-filtering effect and shifts transport toward a mixed regime. A similar trend is observed for the sample annealed at 240 °C for 20 min, as demonstrated in Figure S19. Despite a relatively high W_F of 4.18 eV, the increased $E_F - E_{VBM}$ (0.64 eV) and the suppressed S point to a DOS-scattering-dominated mechanism, as illustrated in the right panel of Figure 4b. At this elevated temperature, thermal degradation and backbone disorder broaden the DOS near the valence-band edge, leading to enhanced carrier scattering and reduced thermoelectric efficiency, even though the apparent W_F remains high. The weak tailed features observed in some spectra originate from second-order/satellite radiation and enhanced inelastic background contributions, which become more apparent after annealing-induced cutoff shifts and do not affect the determination of W_F and E_{VBM}. Overall, the UPS results establish a direct link between interfacial electronic structure and thermoelectric transport in NG/P3HT composites. To clarify the definition of transport mechanism transition, we categorize the transport behavior in the present system based on experimentally observable criteria. In the conventional Pisarenko-dominated regime, the S decreases monotonically with increasing n, and σ follows the classical relation without significant modification of the transport DOS. In contrast, the energy-filtering-assisted regime is identified by a deviation from monotonic S−σ behavior, where S reaches a maximum together with the increased σ. This indicates that interfacial potential fluctuations and DOS reshaping begin to dominate carrier energetics, selectively enhancing high-energy carrier contributions. When structural disorder or excessive interfacial scattering becomes dominant, both σ and S decrease simultaneously, corresponding to a DOS-scattering-limited regime. Therefore, the transport mechanism in the P3HT/NG system is determined by the relative strength of bulk carrier concentration effects and interface-induced DOS modulation, which can be tuned by annealing conditions. The transition from Pisarenko-dominated transport (concentration effect) to energy-filtering-enhanced thermopower (optimal annealing), and finally to DOS-scattering-limited behavior (overannealing), provides a coherent mechanistic framework that explains the observed evolution of Hall and Seebeck parameters. These findings highlight the critical roles of interfacial electronic structure in decoupling thermopower from carrier concentration in organic TE systems.

In summary, we have demonstrated that the thermoelectric transport behavior of P3HT/NG composite films can be systematically modulated by controlling polymer concentration and thermal annealing conditions. Structural characterizations reveal that both increased concentration and moderate annealing enhance the ordering of P3HT on NG, while excessive thermal treatment leads to partial disruption of conjugated segments. Electrical and thermoelectric measurements show that large polymer concentration primarily increases n resulting in Pisarenko-dominated transport with a suppressed S. In contrast, optimized annealing induces a redistribution of the interfacial DOS, as evidenced by UPS, enabling energy-selective carrier transport and a concurrent increase in S and σ. Importantly, the combination of spectroscopic and transport analyses provides direct experimental validation of a transition from carrier concentration-driven to energy-filtering-dominated thermoelectric mechanisms. This work highlights the critical roles of interfacial electronic structure engineering in the OT systems and establishes NG/conjugated polymer hybrids as a versatile platform for breaking the conventional S−σ trade-off.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpclett.6c00487.

Detailed experimental procedures; Raman, optical, XPS, and Hall-effect characterizations of NG; extended Raman and UV−vis spectroscopic analyses of P3HT/NG heterostructures; high-resolution UPS valence band spectra and work-function measurements; and supplementary data supporting the discussion of interfacial electronic modulation in the P3HT/NG hybrid system (PDF)

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The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was financially supported by the National Natural Science Foundation of China (Grant No. 52472044), the Innovation and Development Joint Fund Project of Chongqing Natural Science Foundation (Grant CSTB2023NSCQ-LZX0070), the Natural Science Foundation of Chongqing (Grant No. CSTB2024NSCQ-MSX0755), the Fundamental Research Funds for the Central Universities (Grant No. 2025CDJ-IAISYB-061).

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